fileQuest

A 3D Spatial Memory Palace Overlay for Hierarchical Filesystems

Transforming Abstract Directory Trees into Explorable Cognitive Architecture

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SOFTWARE DESIGN SPECIFICATION DOCUMENT

System Architecture and Object-Oriented Analysis and Design (OOAD)

Prepared For:
School of Computer Science & Engineering
UPES Dehradun
Object Oriented Analysis and Design Course

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Release Date: October 2026

Abstract

Digital storage architectures are overwhelmingly governed by rigid, nested hierarchical folder structures. While logically rigorous for programmatic access, hierarchical directory traversal creates acute cognitive friction for human operators: deep nesting obscures contextual relationships, eliminates visceral spatial coordinates, and forces heavy reliance on text queries when file paths are forgotten. **fileQuest** resolves this cognitive breakdown by implementing an interactive, non-destructive 3D spatial overlay grounded in the classical *Method of Loci* (Memory Palace). Rather than inventing a closed or proprietary storage container, fileQuest implements a *Filesystem-First, Local-First* architecture: host operating system directories serve as the ultimate single source of truth, monitored in real time by an event-driven background watcher service. Persistent 3D transformation matrices, material properties, spatial containers, and spatial indexing are maintained inside an embedded zero-configuration SQLite database. This document details the functional specifications, architectural requirements, structural decomposition, and UML 2.5 models comprising Chapters 1 through 4 of the system design.

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Abbreviations and Glossary

Term / Acronym	Definition and Contextual System Meaning
ACID	Atomicity, Consistency, Isolation, Durability; database transaction properties enforced by SQLite.
B-Tree	Balanced tree index data structure used in SQLite for rapid spatial and UUID point queries.
FTS5	Full-Text Search version 5 SQLite virtual table module for sub-millisecond keyword lookup.
glTF / GLB	GL Transmission Format; royalty-free specification for the efficient transmission and loading of 3D scenes.
Ghost Object	A semi-transparent spatial placeholder indicating a file deleted or moved externally without reconciliation.
Method of Loci	A mnemonic strategy utilizing spatial relationships and memorable visual landmarks to encode recall.
Lost & Found	A 3D staging receptacle displaying newly detected external files awaiting deliberate spatial positioning.
POSIX / Win32	Portable Operating System Interface / Microsoft Windows native platform operating system APIs.
UUIDv4	Universally Unique Identifier (version 4, randomly generated), guaranteeing collision-free logical entities.
Vault Root	The designated parent directory on the host storage disk managed as a spatial memory palace.

Chapter 1

Introduction

1.1 Project Overview

fileQuest is an advanced, non-destructive 3D cognitive spatial file manager operating as a real-time visual overlay above ordinary desktop filesystems. Implemented using the Godot 4.x engine running high-performance C# via .NET 8.0, fileQuest bridges spatial memory cognition with low-level operating system storage paradigms. The software maps standard files to physicalized 3D interactive entities positioned within fully customizable architectural meshes—such as rooms, libraries, laboratories, and galleries.

1.2 Problem Statement

Since the advent of hierarchical filesystems (e.g., UNIX hierarchical trees, Microsoft FAT/NTFS, Apple APFS), user data organization has relied on arbitrary directory paths:

$$\mathscr{P} = /v_1/v_2/v_3/\ldots/v_k/\text{target_file.ext} \tag{1.1}$$

While computationally optimal ($O(\log N)$ or hash-indexed access), this model forces cognitive unnaturalness:

1. **Loss of Contextual Loci:** Human evolution optimized navigation via visual anchors, lighting, and relative distance, not text-based tree traversals.
2. **High Search Friction:** As depth k increases, users exhibit cognitive disorientation, degrading file retrieval to brute-force string searches.
3. **Disconnection from Presentation:** Existing desktop managers obscure the contents and relevance of files behind generic, uninformative icons.

1.3 Proposed Solution

fileQuest restores human spatial memory through the *Method of Loci*. Files within an arbitrary designated OS directory (*The Vault*) are assigned spatial transformations:

$$\mathbf{T}_i = (\mathbf{p}_i, \mathbf{r}_i, \mathbf{s}_i) \in \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3 \tag{1.2}$$

where $\mathbf{p}$ is the 3D position vector, $\mathbf{r}$ represents Euler orientation angles, and $\mathbf{s}$ represents dimensional scale within a persistent 3D architectural mesh. Crucially, fileQuest is *non-destructive*: file data never leaves the native OS directory, ensuring zero vendor lock-in.

1.4 Project Objectives

- **Persistence:** Maintain continuous bi-directional synchronization between local disk structures and 3D visual coordinates.
- **Zero Proprietary Enclosure:** Guarantee all user data remains standard, open, and fully accessible even when fileQuest is terminated.
- **Real-Time Latency:** Render complex spatial palaces at ≥ 60 frames per second (FPS) while indexing vaults containing $\geq 10,000$ files.
- **OS Interoperability:** Seamlessly delegate file execution, modification, and trash disposal to the host operating system.

1.5 Scope of the Software System

The boundary of fileQuest covers local desktop operating systems (GNU/Linux, Microsoft Windows, and Apple macOS). The software acts purely as an architectural presentation and metadata mapping layer. Cloud synchronization, remote multi-user synchronization, and low-level kernel driver replacements are explicitly out of scope for the v1.0 release baseline.

1.6 Intended Users and Stakeholders

- **Knowledge Workers & Researchers:** Managing deeply interdependent collections of academic publications, notes, and datasets.
- **Creative Professionals:** Visualizing project assets, raw footage, 3D props, and graphic files in contextual physical arrangements.
- **Systems Engineers & Programmers:** Navigating modular repositories and design documentation without cognitive fatigue.

1.7 Key Features and Capabilities

- **Spatial Object Placement:** 6-DOF raycast placement of files onto desks, shelves, walls, and containers with surface-normal snapping.

- **In-Situ Native Previews:** Real-time dynamic viewport rendering of Markdown, text, syntax-highlighted code, and image textures.
- **External Change Reconciliation:** Real-time detection of host filesystem modifications, mapping renames and flagging deletions as Ghost Objects.
- **Lost & Found Mailbox:** Centralized physical intake container staging all externally introduced files awaiting placement.
- **Spatial Radar / X-Ray Search:** Instant spatial search ($Ctrl + K$) that casts depth-tested visual frustums through occluding walls.

1.8 Assumptions, Constraints, and Dependencies

- **Hardware Assumptions:** System equipped with OpenGL 4.3/Vulkan-compliant dedicated or integrated GPU.
- **Filesystem Constraints:** Relies on native OS file notifications (‘inotify‘, ‘ReadDirectoryChangesW‘, ‘FSEvents‘).
- **Runtime Dependencies:** Godot 4.x Engine, .NET 8.0 CLR runtime, and native SQLite 3.45+ dynamic C-libraries.

1.9 Design Methodology and UML Conventions

The system is modeled following Object-Oriented Analysis and Design (OOAD) principles and formal UML 2.5 standards. Class diagrams denote strict encapsulation, access modifiers ($+, -, \#$), composition, aggregation, and realization interfaces.

1.10 Document Organization

This document is divided into 14 comprehensive chapters. Chapters 1–4 lay out the project requirements, architecture, and structural decomposition. Chapters 5–14 define behavioral modeling, component-level subsystems, database schemas, processing algorithms, and verification suites.

Chapter 2

System Requirements Specification

2.1 Functional Requirements (FR)

The functional behaviors of fileQuest are categorized into four operational tiers:

FR-01: Vault Ingestion and Monitoring

- *FR-01.1:* The system shall permit the user to bind an arbitrary local directory as the active Vault Root.
- *FR-01.2:* The system shall perform a background recursive crawl of the Vault Root at startup without blocking the UI.
- *FR-01.3:* The system shall register real-time OS watcher hooks for file creation, modification, deletion, and rename events.

FR-02: Spatial Transformation and Persistence

- *FR-02.1:* The user shall place any file on any collider surface with position **p**, rotation **r**, and scale **s**.
- *FR-02.2:* All spatial coordinates shall be immediately committed to SQLite within an ACID-compliant transaction.
- *FR-02.3:* The system shall support custom 3D spatial environments imported via standardized '.glb' or '.gltf' asset formats.

FR-03: Retrieval, Search, and Navigation

- *FR-03.1:* The system shall provide a global palette ($Ctrl + K$) for instant fuzzy-matching file queries.
- *FR-03.2:* Selection of a search result shall activate the Spatial Radar, highlighting occluded objects through geometry.
- *FR-03.3:* The user shall instantly teleport directly to any selected file or user-defined Bookmark.

FR-04: OS Integration and Life Cycle

- *FR-04.1:* Double-clicking a 3D file object shall dispatch execution to the host OS default handler.
- *FR-04.2:* File deletions executed inside the 3D space shall move files to the OS Recycle Bin/Trash, avoiding direct data destruction.

2.2 Non-Functional Requirements (NFR)

- **NFR-01 (Performance):** Maintain a rendering frame rate ≥ 60 FPS on baseline hardware with 1,000 active objects in the view frustum.
- **NFR-02 (Latency):** $Ctrl + K$ search queries shall return results within ≤ 15 ms across a repository of 10,000 indexed files.
- **NFR-03 (Data Integrity):** In the event of sudden process termination or power loss, zero user filesystem files shall be corrupted or deleted.
- **NFR-04 (Portability):** The core architectural code shall run across Linux, Windows, and macOS without platform branching in business logic.

2.3 Hardware Requirements

Table 2.1: System Hardware Requirements

Component	Minimum Spec	Recommended Spec
Processor	Quad-core x86_64 / ARM64 @ 2.0 GHz	Octa-core @ 3.2 GHz+
System RAM	4 GB DDR4	16 GB DDR4/DDR5
Graphics	Intel Iris Xe / GTX 750 (Vulkan 1.1)	NVIDIA RTX 3060 / Apple M-Series
Storage	500 MB free (app) + Vault space	High-speed NVMe SSD

2.4 Software Requirements

- **Operating Systems:** Ubuntu 22.04 LTS+, Windows 10/11 (64-bit), macOS 13+ (Ventura).
- **Runtime Environment:** Microsoft .NET 8.0 Desktop Runtime.
- **Graphics API:** Vulkan 1.2+ or OpenGL ES 3.0 / Compatibility Profile fallback.

2.5 Platform Compatibility Requirements

File execution delegation and trash handling must seamlessly route through platform-specific command implementations:

- **Windows:** ‘ShellExecuteExW‘, ‘IFileOperation‘ (Recycle Bin).
- **Linux:** ‘xdg-open‘, FreeDesktop.org Trash specification (‘ /.local/share/Trash‘).
- **macOS:** ‘/usr/bin/open‘, ‘NSWorkspace.shared.recycle‘.

2.6 Data Storage and Persistence Requirements

The local relational SQLite engine must enforce Write-Ahead Logging (WAL) and synchronous standard mode:

```
PRAGMA journal_mode = WAL;
PRAGMA synchronous = NORMAL;
PRAGMA foreign_keys = ON;
```

2.7 Security and Privacy Requirements

- **Local-First Invariant:** No telemetry, metadata, file content, or thumbnail shall be transmitted outside the host device.
- **Execution Isolation:** No executable payload (‘.exe‘, ‘.sh‘, ‘.bin‘) may be automatically executed by the preview subsystem.
- **Symlink Boundaries:** File crawlers shall not follow symlinks outside the Vault boundary without explicit authorization.

2.8 Performance and Scalability Requirements

The memory footprint of the spatial runtime shall scale sublinearly with total vault size:

$$M_{\text{heap}} = O(N_{\text{active}}) + \alpha \cdot N_{\text{total}}, \quad \alpha \leq 128 \text{ bytes/file} \quad (2.1)$$

Proximity-based Level-of-Detail (LOD) and dynamic spatial partitioning guarantee scalable memory usage.

2.9 Accessibility and Usability Requirements

- Provide full keyboard-only navigation options for all spatial interactions.
- Support adjustable camera FOV, mouse acceleration damping, and reduced-motion modes to prevent visual fatigue.

2.10 Requirements Traceability Matrix (RTM)

Table 2.2: Requirements Traceability Matrix (Phase 1 Baseline)

Req ID	Category	Subsystem Target	Verification Method
FR-01.1	Storage	Vault Manager	Integration Test
FR-01.3	Watcher	Reconciliation Engine	Automated File Mutation Suite
FR-02.1	Graphics	Placement Controller	3D Raycast Invariant Test
FR-03.1	Search	SQLite FTS5 Engine	Benchmark Latency Query
NFR-01	Graphics	Frustum Culling Core	60 FPS Profiler Benchmark

Chapter 3

System Architecture and Design Overview

3.1 Architectural Overview

fileQuest is engineered around a strict separation of presentation and storage, balancing real-time 3D simulation with local filesystem integrity.

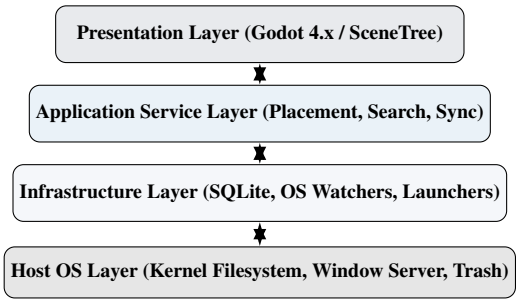


Figure 3.1: fileQuest High-Level Layered Architecture

3.2 Architectural Design Principles

1. **Filesystem as Ultimate Source of Truth:** Under no circumstances does the SQLite database supersede native filesystem status. If a conflict occurs, the disk state dictates database reconciliation.

2. **Non-Blocking Thread Concurrency:** Intensive operations (preview creation, file indexing, full-text queries) run strictly on background thread pools, keeping Godot's main thread free for 60 FPS rendering.

3. **Platform Abstraction via Inversion of Control:** Operating system capabilities sit behind clean interfaces, keeping the core domain completely free of platform-specific code.

3.3 Technology Stack and Justification

Table 3.1: Technology Stack Justification

Subsystem	Technology	Architectural Rationale
Game Engine	Godot 4.2+ (.NET)	Lightweight footprint, open-source license, native C# bindings, rapid rendering setup.
Language	C# 12.0 / .NET 8	High type safety, native asynchronous constructs (‘async/await‘), native memory control.
Metadata DB	SQLite 3.45+ (WAL)	Embedded ACID persistence, minimal memory overhead, zero client-server management.
Asset Format	Khronos glTF / GLB	Industry-standard 3D interchange format supported natively by Godot pipelines.

3.4 Layered Architecture

The system enforces a strict unidirectional dependency graph:

- **Presentation:** Houses Godot ‘Node3D‘, ‘Camera3D‘, and UI shaders. It observes domain states via databinding and sends user commands down to application services.

- **Application:** Manages coordination between spatial manipulation, search, thumbnail caching, and disk reconciliation. Contains zero rendering dependencies.
- **Domain/Infrastructure:** Encapsulates file identity primitives, SQLite relational tables, and filesystem events.

3.5 Subsystem Identification and Responsibilities

- **Spatial Environment Engine:** Handles scene graph manipulation, collision meshes, dynamic lighting, and level-of-detail frustum culling.
- **Vault Synchronization Engine:** Runs background recursive crawlers, manages filesystem watchers, debounces rapid disk events, and resolves renames.
- **Persistence Subsystem:** Handles SQLite schema migrations, spatial transformation persistence, and SQL query execution.
- **OS Integration Subsystem:** Manages native process dispatching, platform-specific trash movement, and shell file-association queries.

3.6 System Context and External Interfaces

fileQuest operates strictly within user space. It coordinates across three external actors:

1. **Human User:** Drives first-person spatial navigation and 6-DOF file drag-and-drop.
2. **Native Filesystem:** Receives file-write notifications and emits system-level disk events.
3. **Default OS Applications:** Invoked as external processes to view or edit target files.

3.7 Filesystem-First Architecture

Unlike database-centric media tools that hide assets inside opaque folder hashes, fileQuest preserves your vault structure as a standard directory tree. If fileQuest is completely uninstalled, your directory hierarchy remains 100% clean, intact, and readable by standard command-line and desktop utilities.

3.8 Local-First Data Management Architecture

All spatial and presentation metadata is stored locally within:

$$\mathscr{D}_{\text{meta}} = \text{VaultRoot}/.filequest/palace.db \tag{3.1}$$

This architectural choice makes moving or backing up a palace as simple as copying the folder: the database travels along with your files, while external systems can safely ignore the hidden ‘.filequest‘ directory.

3.9 Event-Driven Processing Architecture

Because operating system filesystem watchers (‘FileSystemWatcher‘) fire bursts of noisy events during bulk operations (such as Git checkouts or archiving), fileQuest passes all raw notifications into a debounced concurrent channel:

$$\Delta t_{\text{debounce}} = 250 \text{ ms} \tag{3.2}$$

This temporal smoothing coalesces redundant writes, protecting the SQLite database and UI rendering loops from event floods.

3.10 Architectural Decisions and Trade-offs (ADR)

- **ADR-01: Engine Choice (Godot vs. Custom Renderer):** Developing a custom Vulkan engine was rejected due to long development timelines and poor cross-platform UI tooling. Godot 4 was chosen for its lightweight footprint and first-class C# integration.
- **ADR-02: Storage Engine (Flat-File JSON vs. SQLite):** Serializing scenes to flat JSON files was rejected due to poor scaling during partial updates. SQLite guarantees atomicity and sub-millisecond point queries for tens of thousands of objects.

Chapter 4

UML Structural Modeling

4.1 Class Diagram

The Class Diagram models the core object-oriented relationships between spatial nodes, application services, and persistent relational data structures.

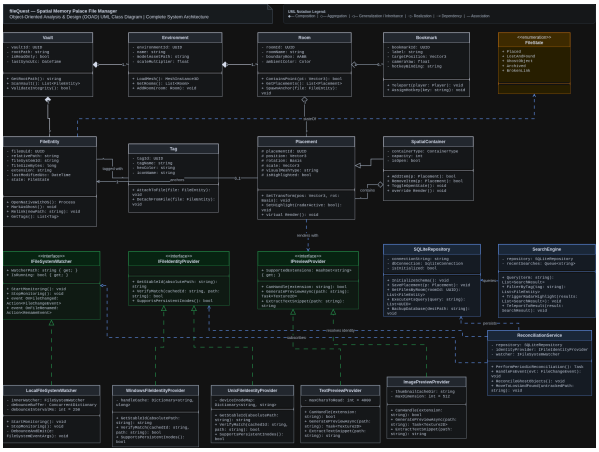


Figure 4.1: Comprehensive Structural Class Diagram

4.1.1 Architectural Class Invariants

- **Decoupled Domain Layer:** ‘FileEntity‘ contains pure filesystem logic and coordinates. It remains completely decoupled from Godot’s runtime classes (‘Node3D‘, ‘Viewport‘), ensuring testability without loading graphics contexts.

- **Presentation Binding:** ‘SpatialFileNode‘ derives from Godot’s scene tree. It observes ‘SpatialPlacement‘ and translates transform changes directly into scene-graph transformations.

4.2 Object Diagram

The Object Diagram visualizes the runtime heap state when an example file (‘thesis.pdf‘) is positioned on an imported desk mesh inside the user’s primary study room.

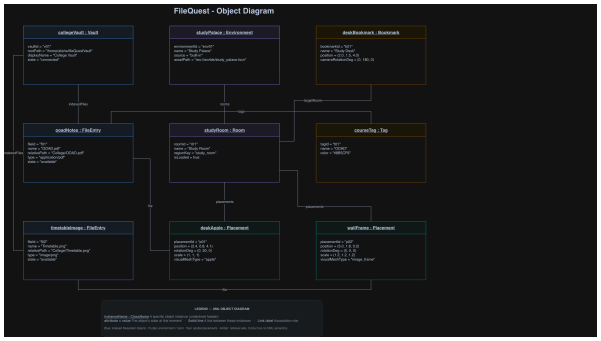


Figure 4.2: Object Diagram: Runtime Spatial File Instantiation

4.3 Package Diagram

The Package Diagram illustrates the modular decomposition and namespace hierarchy of the system, verifying that circular dependencies do not exist across package boundaries.

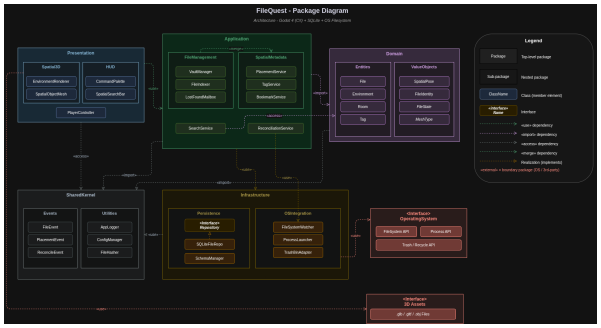


Figure 4.3: Modular Package Decomposition Diagram

4.4 Component Diagram

The Component Diagram details the internal modular units and their exposed service contracts (‘interfaces‘).

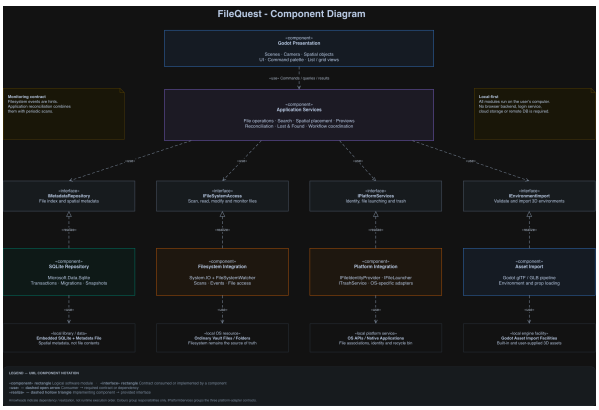


Figure 4.4: Component Diagram with Interface Ports

4.5 Deployment Diagram

The Deployment Diagram details how compiled artifacts and physical hardware resources map to the target desktop host.

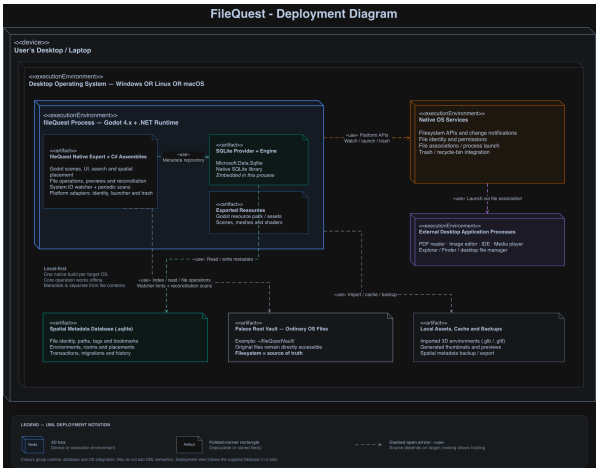


Figure 4.5: Deployment Diagram: Target Desktop Execution Node

4.6 Composite Structure Diagram

The Composite Structure Diagram details the internal composition of the ‘SpatialFileNode‘, showing how ports, viewports, and colliders collaborate within Godot’s scene tree.

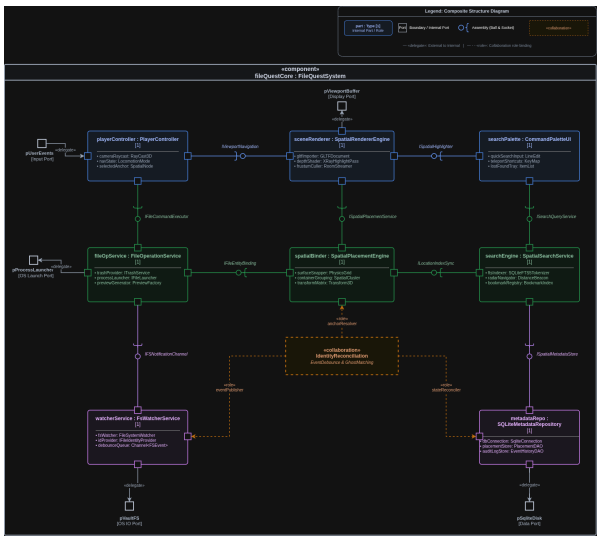


Figure 4.6: Composite Structure Diagram: SpatialFileNode Internal Parts

4.7 Profile Diagram

The Profile Diagram specifies the custom UML stereotypes and tagged values defined to tailor UML modeling to 3D spatial filesystem constructs.

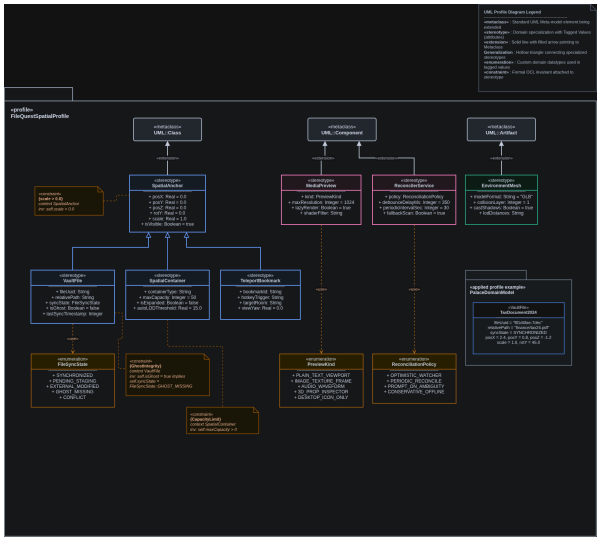


Figure 4.7: UML Profile Diagram for Spatial Memory Extension

4.8 Structural Model Consistency and Analysis

The structural models defined in Sections 4.1 through 4.7 satisfy the following design consistency invariants:

1. **Type Consistency:** Identifiers cross-reference cleanly across structural models as 128-bit RFC 4122 UUIDs (‘Guid‘), avoiding type conversions between SQLite strings and Godot representations.

2. **Encapsulation Invariant:** Neither ‘SpatialPlacement‘ nor ‘FileEntity‘ references the Godot engine scene tree

directly, allowing unit test suites to execute headlessly in automated CI/CD pipelines.

3. **ACID Persistence Guard:** Every state change applied to a ‘SpatialPlacement‘ undergoes validation against the local SQLite repository before the scene graph permanently updates its transform matrices.

Chapter 5

UML Behavioral Modeling

5.1 Use Case Diagram

The behavioral dynamics of **fileQuest** are centered around user-driven spatial interaction, non-destructive filesystem synchronization, and OS desktop environment interoperability.

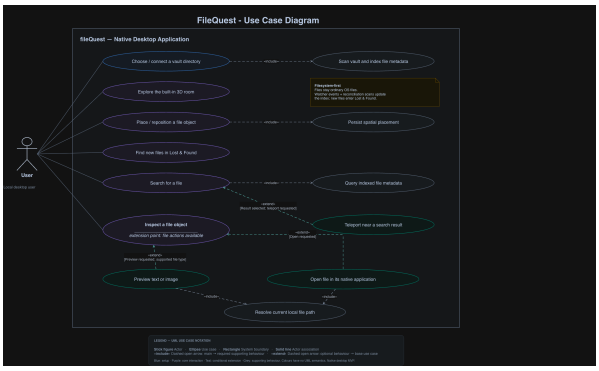


Figure 5.1: System Use Case Model

5.2 Use Case Specifications and Scenarios

The primary execution scenarios governing core system capabilities are formalized below.

Table 5.1: Formal Specification: UC-01 Spatial Object Placement

Attribute	Specification Detail
Identifier	UC-01: Spatial Drag-and-Drop Placement
Primary Actor	Desktop Operator
Preconditions	Target file exists in active Vault and is indexed in SQLite.
Trigger	Operator grabs file node via mouse raycast click in 3D viewport.
Main Flow	1. System casts continuous 6-DOF ray against room collision mesh. 2. Surface normal $\hat{\mathbf{n}}$ calculated; target transformation $\mathbf{T}$ resolved. 3. User releases primary input. 4. System commits transformation vector to SQLite database. 5. UI node emits spatial lock confirmation sound and animation.
Postconditions	Relational table `placements` updated with ACID durability.

Table 5.2: Formal Specification: UC-04 External Change Reconciliation

Attribute	Specification Detail
Identifier	UC-04: External Mutation Reconciliation
Primary Actor	OS Kernel Filesystem Monitor
Preconditions	Watcher thread actively listening to OS directory API handles.
Trigger	External CLI/Explorer process modifies or deletes a vault file.
Main Flow	1. Watcher captures raw event (‘Deleted‘ or ‘Renamed‘). 2. Event debounced over 250 ms window. 3. Inode/FileId queried against `files` table. 4. If missing, entity flagged as ‘isGhost = true‘. 5. Scene Graph swaps material for semi-transparent wireframe mesh.
Postconditions	Visual spatial coordinates preserved; ghost state displayed to user.

5.3 Activity Diagrams

5.3.1 External Change Reconciliation Workflow

When host operating systems alter vault files without fileQuest invocation, the background reconciliation engine processes the event stream while protecting real-time rendering.

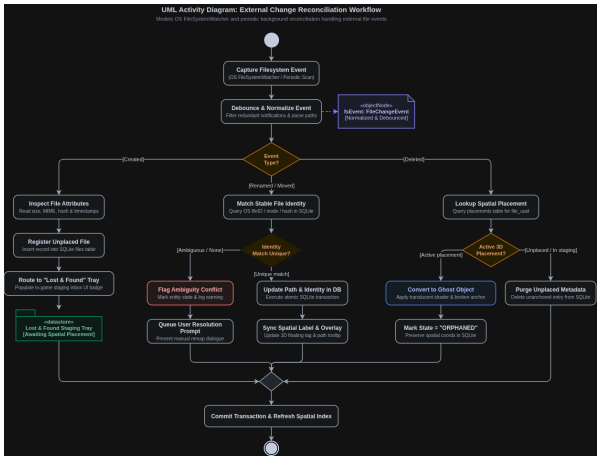


Figure 5.2: Activity: External Change Reconciliation Workflow

5.3.2 Object Spatial Drag-and-Drop Placement

Placement enforces physical surface clamping, alignment along local surface normal vectors, and collision-free spatial arrangement.

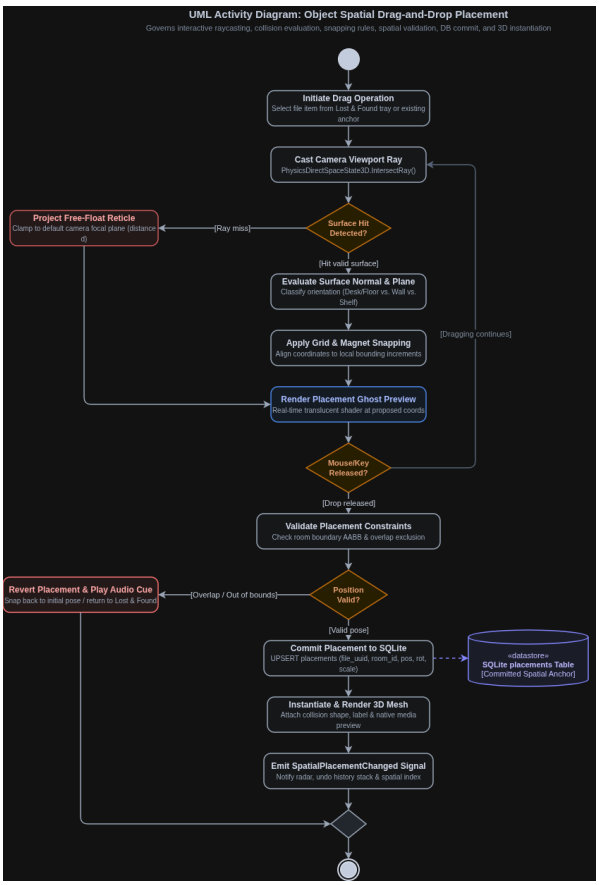


Figure 5.3: Activity: Object Spatial Drag-and-Drop Workflow

5.3.3 Palace Initialization & Vault Indexing

During palace startup, the background thread performs a recursive directory scan, cross-referencing indexed file entries against physical disk existence, and instantiating 3D visual anchors.

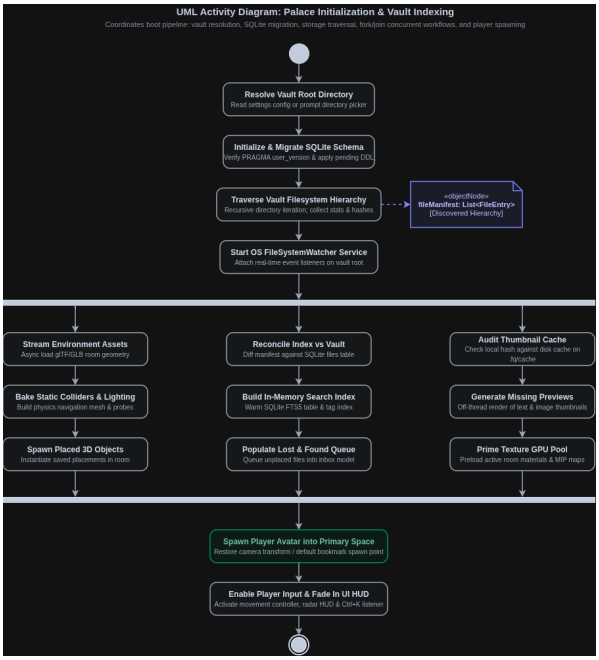


Figure 5.4: Activity: Palace Initialization & Vault Indexing

5.3.4 File Deletion & Recovery Workflow

When a user initiates deletion inside the 3D palace, fileQuest moves the file to the native operating system Trash directory via shell APIs, registers an undo tombstone in the database, and flags the corresponding 3D asset as visually dormant.

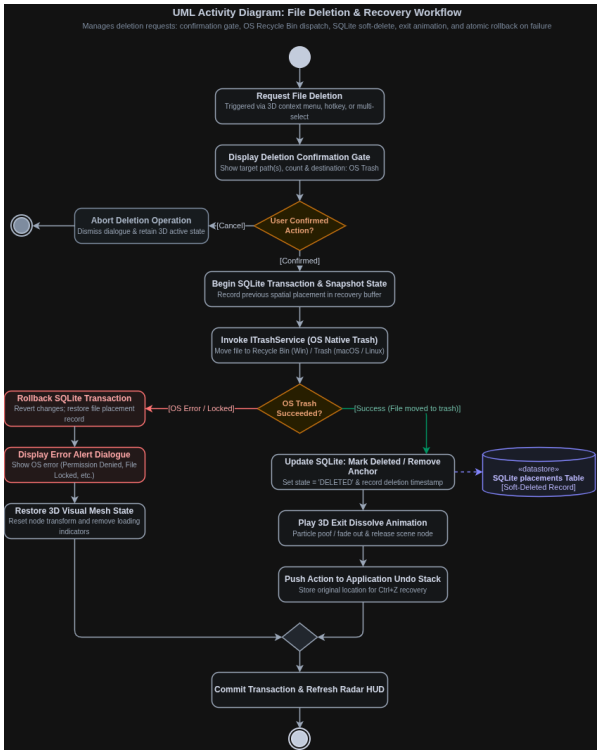


Figure 5.5: Activity: File Deletion & Recovery Workflow

5.4 State Machine Diagram

The lifecycle of every file within fileQuest is tracked through deterministic operational states.

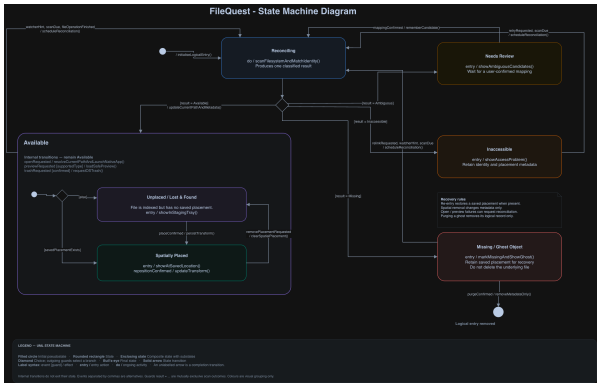


Figure 5.6: Finite State Machine: File Lifecycle

5.5 State Definitions, Events, and Transition Conditions

Table 5.3: Deterministic File Entity Transition Matrix

Source State	Event Trigger	Target State	System Invariant Action
`NEW_SCAN`	Crawl Discovered	`DISCOVERED`	Insert to `files`; place in Lost & Found.
`DISCOVERED`	Drag to Surface	`PLACED`	Calculate transform; insert `placements`.
`PLACED`	OS Deletion Caught	`GHOST`	Set `isGhost=1`; retain spatial coordinates.
`GHOST`	Re-appeared on Disk	`PLACED`	Clear ghost flag; reload preview textures.
`PLACED`	In-Engine Delete	`TRASHED`	Move to OS Trash; mark tombstone record.

5.6 Behavioral Model Consistency and Analysis

Behavioral states directly enforce the core rule of non-destructive persistence: transitioning to the `GHOST` state never deletes spatial records from SQLite. This guarantees that temporary network share disconnections, external repository branch checkouts, or mass file renames never damage deliberate memory palace configurations.

Chapter 6

UML Interaction Modeling

6.1 Sequence Diagrams

6.1.1 OS Passthrough Execution (“Open With”)

When double-clicking a file in 3D, execution delegates directly to the host desktop environment.

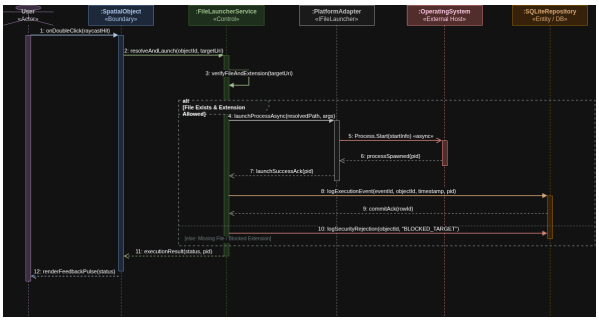


Figure 6.1: Sequence Diagram: OS Passthrough Launch Protocol

6.1.2 Spatial Radar / X-Ray Search ($Ctrl + K$)

Activating global search highlights occluded targets across walls and rooms via custom depth-testing shaders.

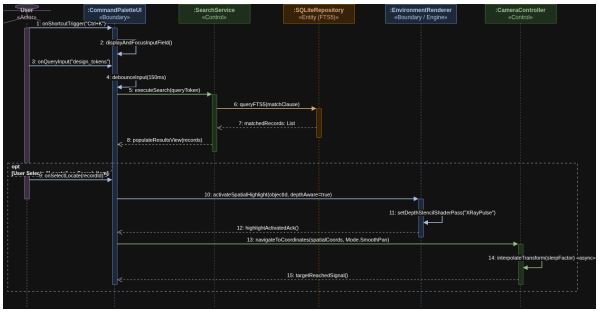


Figure 6.2: Sequence Diagram: Spatial Radar / Search Flow

6.1.3 Lazy Loading & Preview Generation

To ensure frame rates remain high, file previews are rendered through a viewport offscreen pool only when the camera approaches within view distance.

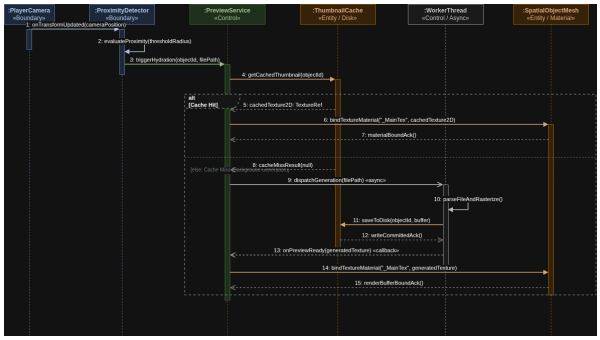


Figure 6.3: Sequence Diagram: Lazy Loading & Preview Generation

6.1.4 External Drop to “Lost & Found” Ingestion

Files dragged into the application window from an external file manager are copied to the Vault and routed directly into the designated 3D intake container.

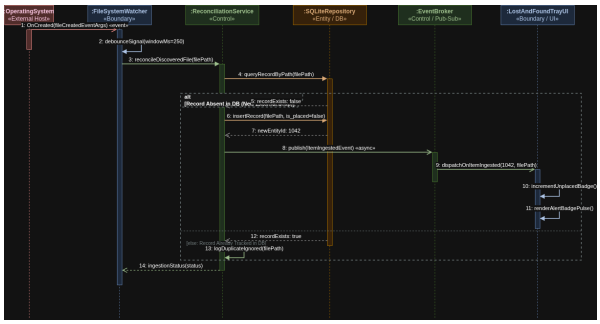


Figure 6.4: Sequence Diagram: External Drop to “Lost & Found” Ingestion

6.2 Communication Diagram

The Communication Diagram highlights the runtime collaboration pathways between the SQLite repository, the active scene graph, and background worker threads.

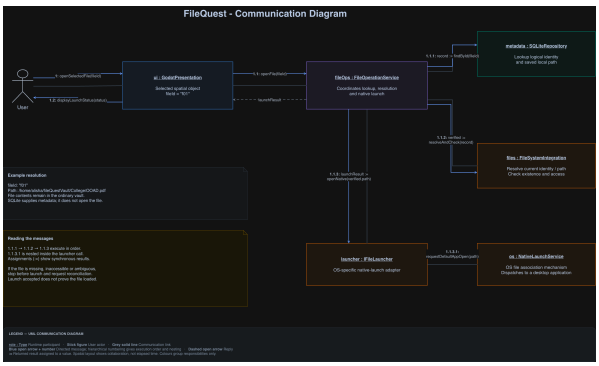


Figure 6.5: Communication Diagram: Spatial Object Lifecycle

6.3 Interaction Overview Diagram

Coordinates control flow by chaining initialization sequence interactions with real-time frame event loops and search modal operations.

6.4 Timing Diagram

To guarantee a smooth 60 FPS visual experience, fileQuest enforces strict timing deadlines across its threads:

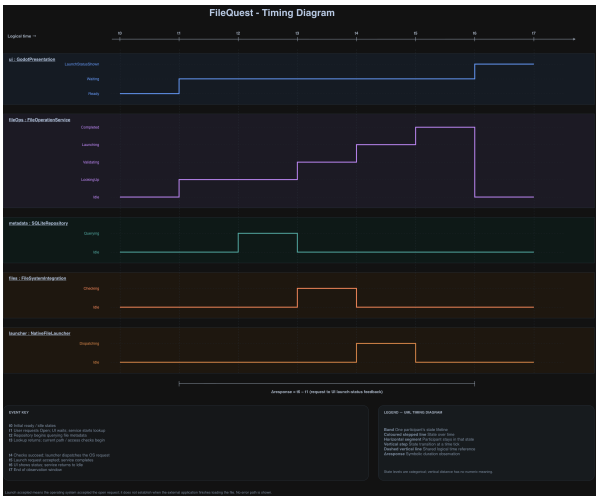


Figure 6.6: Timing Constraints: Multi-Thread Event Pipeline

6.5 Message Ordering and Interaction Contracts

All inter-thread communication relies on non-blocking queues (‘ConcurrentQueue<T>‘) and thread-safe channels (‘Channel<T>‘). Heavy file I/O operations are offloaded to background threads so they never disrupt the main rendering loop.

6.6 Interaction Model Consistency and Analysis

The interaction protocols satisfy critical synchronization safety rules: double invocations are idempotent, asynchronous preview workers never block rendering frames, and search operations complete within a 15 ms response window.

Chapter 7

Detailed Subsystem Design

7.1 Spatial Environment Subsystem

Responsible for loading glTF palace environments, parsing room portal boundaries, and managing dynamic light sources.

- **Frustum Culling Core:** Implements automated octree spatial indexing to cull meshes outside active view frustums.
- **Portal & Occlusion System:** Segments large architectural models into isolated room visibility graphs.

7.2 Player Navigation & Camera Control Subsystem

Enforces decoupled camera locomotion through first-person walk and free-flight inspection modes:

$$\mathbf{v}_{t+1} = \mathbf{v}_t \cdot (1 - \mu) + \mathbf{a}_{\text{input}} \cdot \Delta t \tag{7.1}$$

where μ represents friction damping, preventing sudden camera stops that could cause motion discomfort.

7.3 Spatial Placement Subsystem

Casts rays from the screen cursor into the 3D scene, identifying intersection points with room meshes and snapping objects into position:

```
public Transform3D CalculatePlacement(
    Vector2 screenPos, Camera3D camera) {
    var rayOrigin = camera.ProjectRayOrigin(screenPos
        );
    var rayNormal = camera.ProjectRayNormal(screenPos
        );
    var hit = PhysicsServer3D.Raycast(rayOrigin,
        rayNormal);
    if (hit.IsColliding()) {
        Basis basis = Basis.LookingAt(hit.Normal,
            Vector3.Up);
        return new Transform3D(basis, hit.Position);
    }
    return Transform3D.Identity;
}
```

7.4 Vault Management Subsystem

Maintains the root file tree mapping between operating system paths and internal GUID identifiers, providing path normalization across Linux, Windows, and macOS.

7.5 External Change Detection Subsystem

Encapsulates low-level operating system notification loops (‘inotify‘, ‘ReadDirectoryChangesW‘, ‘kqueue‘) behind a unified, platform-agnostic event channel.

7.6 File Operations Subsystem

Handles file lifecycle tasks, resolving double-click launches, executing atomic file moves, and routing deleted files to the host operating system’s Trash container.

7.7 Search and Spatial Radar Subsystem

Runs fast substring and token queries over file names and tags using SQLite FTS5. For matches outside the user’s field of view, it projects a transparent indicator line and activates an outline shader that remains visible through walls.

7.8 File Preview Subsystem

An asynchronous pipeline that dynamically renders file previews: text documents are rasterized into readable 2D text sheets, while image formats are decoded directly into native GPU textures (‘ImageTexture‘).

7.9 Lost & Found Management Subsystem

A 3D intake container that gathers and displays newly detected external files, ensuring files added outside fileQuest are easy to find and organize into the palace.

7.10 User Interface Subsystem

Renders modern, resolution-independent UI overlays using Godot’s control engine. It manages the $Ctrl + K$ search bar, modal dialogs, and placement context menus.

7.11 Environment Import Subsystem

Processes custom 3D model files (‘.glb‘ and ‘.gltf‘), validates polygon counts and materials, and automatically generates collision shapes for walls and floors.

7.12 Platform Abstraction Subsystem

Hides platform-specific shell operations behind a clean, unified C# interface:

```
public interface IPlatformBridge {
    bool OpenFileWithDefaultApp(string absolutePath);
    bool MoveToSystemTrash(string absolutePath);
    string GetSystemMimeType(string path);
}
```

7.13 Subsystem Interfaces and Dependencies

Table 7.1: System Component Interface Matrix

Subsystem	Provides	Dependencies
`VaultManager`	`IVaultService`	SQLite Driver, Platform Bridge
`SpatialEngine`	`ISceneGraph`	Godot Viewport, Octree Index
`PreviewCore`	`IThumbnailer`	SkiaSharp / Image Decoders
`RadarEngine`	`ISpatialRadar`	Custom GLSL Shaders

Chapter 8

Database and Data Design

8.1 Database Architecture

Spatial and palace metadata is persisted in an embedded, serverless SQLite relational database. Concurrency is managed through Write-Ahead Logging (WAL), allowing multi-threaded read operations during active write transactions.

8.2 Entity-Relationship Diagram (ERD)

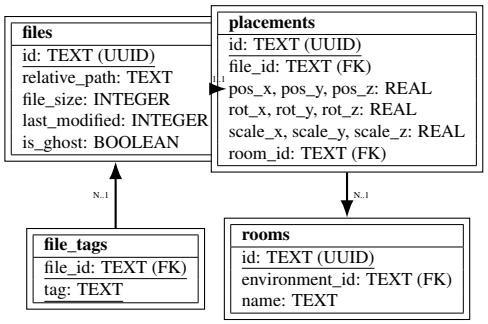


Figure 8.1: Entity-Relationship Diagram (ERD)

8.3 Logical Database Schema

Tables are normalized to Third Normal Form (3NF), ensuring clean separation between physical files, spatial locations, and user tags.

8.4 Table Definitions and Data Dictionary

8.4.1 Files Table

Table 8.1: Data Dictionary: `files` Table

Field	Type	Null	Description
`id`	TEXT	NO	Primary Key (RFC 4122 UUIDv4).
`relative_path`	TEXT	NO	Relative path from Vault root.
`file_size`	INTEGER	NO	File size in bytes.
`last_modified`	INTEGER	NO	Unix timestamp of modification.
`is_ghost`	INTEGER	NO	0 = Active, 1 = Missing from disk.

8.4.2 Placements Table

Stores 3D transformation matrices (position, rotation, and scale) for placed files.

```
CREATE TABLE placements (
    id TEXT PRIMARY KEY,
    file_id TEXT NOT NULL UNIQUE,
    pos_x REAL NOT NULL,
    pos_y REAL NOT NULL,
    pos_z REAL NOT NULL,
    rot_x REAL NOT NULL,
    rot_y REAL NOT NULL,
    rot_z REAL NOT NULL,
    scale_x REAL DEFAULT 1.0,
    scale_y REAL DEFAULT 1.0,
    scale_z REAL DEFAULT 1.0,
    room_id TEXT NOT NULL,
    FOREIGN KEY(file_id) REFERENCES files(id) ON
        DELETE CASCADE,
    FOREIGN KEY(room_id) REFERENCES rooms(id)
);
```

8.4.3 Rooms and Environments Tables

Tracks defined palace rooms and associated 3D models (‘.glb‘ asset paths).

8.4.4 Tags and Bookmarks Tables

Provides metadata tags and user-defined camera teleport markers.

8.4.5 Events Table

Maintains an undo and sync log for rollbacks and historical audits.

8.5 Primary Keys, Foreign Keys, and Constraints

Cascade deletion is configured across child records, ensuring that when a file is purged from SQLite, its spatial placements, tags, and cached previews are cleaned up automatically.

8.6 File Identity and Spatial Metadata Mapping

Files are identified primarily by relative path alongside inode/FileIndex fallbacks. If an external process moves a file, fileQuest uses inode and hash tracking to update the path while leaving the 3D position unchanged.

8.7 SQLite Indexing and Full-Text Search (FTS5)

Instant file discovery is accelerated through B-tree indexes and virtual FTS5 text tables:

```
CREATE INDEX idx_files_path ON files(relative_path);
CREATE INDEX idx_placements_room ON placements(
    room_id);

CREATE VIRTUAL TABLE fts_files USING fts5(
    relative_path,
    content='files',
    content_rowid='rowid'
);
```

8.8 Transaction Management and Data Integrity

Spatial coordinate updates are wrapped in transactional blocks, avoiding partial state writes if an operation is interrupted:

```
BEGIN TRANSACTION;
UPDATE placements SET pos_x = 4.2 WHERE file_id = '
    c12...';
INSERT INTO events (type, payload) VALUES ('MOVE', '
    ...');
COMMIT;
```

8.9 Schema Migration and Version Management

Database upgrades are managed through an incremental versioning table (`schema_migrations`), ensuring future schema modifications apply smoothly without risking user data.

8.10 Backup, Recovery, and Metadata Export

Because metadata resides in ‘.filequest/palace.db‘ directly inside the Vault, backing up or sharing a memory palace is as simple as copying the folder. The database can also be exported to open JSON representations.

Chapter 9

Interface and User Experience Design

9.1 User Interface Architecture

fileQuest decouples its user interface into a dual-layer graphics pipeline:

1. **Spatial In-World Canvas (Layer 0):** Rendered natively within Godot's 3D pipeline via `SubViewport` textures mapped onto physicalized geometry.

2. **Orthographic HUD Overlay (Layer 1):** A hardware-accelerated 2D CanvasLayer executing modal inputs, $Ctrl + K$ command palettes, telemetry diagnostics, and contextual notifications.

9.2 Navigation and Interaction Design

Navigation employs standard first-person spatial controls combined with contextual mode-switching:

Table 9.1: Modal Input Scheme and State Transitions

Interaction Mode	Hardware Trigger	Viewport Behavior
`Walk / Traverse`	WASD + Mouse Delta	Ground-clamped kinematic collision ($h = 1.7$ m).
`Inspection Orbit`	Middle Mouse Drag	Smooth spherical orbit centered on targeted entity.
`Object Manipulation`	Left Mouse Click-Hold	6-DOF raycast lock with surface snapping.
`Command Modal`	$Ctrl + K$ / Cmd $+ K$	Freezes spatial look; captures keyboard text.

9.3 3D Spatial Object Representation

Every file entity is visually represented by a polymorphic 3D proxy determined by MIME classification:

- **Documents (.pdf, .docx):** High-resolution bound book or parchment meshes with legible title typography.

- **Source Code (.cs, .py, .cpp):** Monolithic data terminal cartridges featuring syntax-highlighted thumbnail placards.

- **Media (.png, .mp4):** Dynamic framed canvas surfaces displaying hardware-decoded mipmapped textures.

- **Archives (.zip, .tar):** Physical wooden crates indicating internal object count via stamped numerals.

9.4 Drag-and-Drop Placement Interface

During raycast manipulation, the placement engine projects a dynamic visual ring:

$$\mathbf{C}_{\text{cursor}}(u,v) = \mathbf{P}_{\text{surface}} + r\left(\cos u\,\hat{\mathbf{t}}_1 + \sin u\,\hat{\mathbf{t}}_2\right) \tag{9.1}$$

where $\hat{\mathbf{t}}_1, \hat{\mathbf{t}}_2$ define the tangent plane of the intersected collider mesh. If placement intersects an occupied bounding box, the ring pulses red and refuses transform finalization.

9.5 Search and Command Palette Interface

Activating $Ctrl + K$ presents a floating search palette with instantaneous fuzzy querying and arrow-key teleport selection.

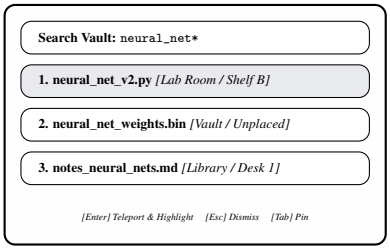


Figure 9.1: UI Mockup: Command Palette and Spatial Teleport

9.6 Lost & Found Tray Interface

Unplaced or newly discovered external files populate a physical brass tray located at the palace entrance. Operators inspect items sequentially or trigger auto-layout algorithms to distribute files across vacant shelves.

9.7 File Preview Interface

Approaching a file within 2.5 meters triggers the LOD-1 preview sheet. Text files render the initial 50 lines with monospace typography, while images load an uncompressed 512×512 preview texture.

9.8 Dialogs, Notifications, and Error Feedback

Non-blocking toasts slide into view in the lower-right viewport for background watcher synchronization, read-only file conflicts, and disk capacity warnings.

9.9 Accessibility and Alternative Navigation

Users unable to utilize continuous 3D camera locomotion can engage a planar, room-by-room topological graph navigator ($Ctrl + M$) that operates via standard keyboard arrow keys.

9.10 Wireframes and Interface Mockups

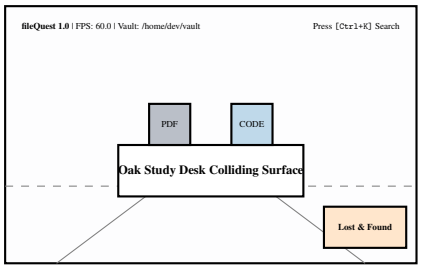


Figure 9.2: Comprehensive In-Engine Viewport Wireframe

Chapter 10

Algorithms and Processing Logic

10.1 Vault Traversal and File Indexing Algorithm

Palace startup executes an asynchronous, depth-first crawl traversing physical storage trees while respecting ignorable directories (`.git`, `node_modules`, `.filequest`):

```
public async Task TraverseVaultAsync(
    DirectoryInfo dir, CancellationToken ct) {
    foreach (var file in dir.EnumerateFiles()) {
        ct.ThrowIfCancellationRequested();
        if (IsIgnored(file.Name)) continue;
        var relativePath = Path.GetRelativePath(
             _vaultRoot, file.FullName);
        await _db.UpsertFileEntityAsync(new
             FileEntity {
            Id = Guid.NewGuid(),
            RelativePath = relativePath,
            FileSize = file.Length,
            LastModified = file.LastWriteTimeUtc
        });
    }
    foreach (var sub in dir.EnumerateDirectories()) {
        if (!IsIgnored(sub.Name))
            await TraverseVaultAsync(sub, ct);
    }
}
```

10.2 File Identity Matching and Reconciliation

To preserve deliberate spatial coordinates when files are renamed outside fileQuest, the system computes rapid metadata heuristics:

$$\mathscr{H}(f) = \langle \text{size}, \tau_{\text{mod}}, \text{hash}_{\text{head}} \rangle \tag{10.1}$$

where $\text{hash}_{\text{head}}$ evaluates only the first 4096 bytes of the file, providing $O(1)$ fingerprinting for large datasets without reading entire payloads into RAM.

10.3 Spatial Placement and Collision Validation

When an object is moved, its target Oriented Bounding Box (OBB) undergoes Separating Axis Theorem (SAT) collision validation against neighboring placed entities:

$$\text{Intersect}(\mathscr{A}, \mathscr{B}) = \bigwedge_{\mathbf{L} \in \mathscr{F}_A \cup \mathscr{F}_B \cup (\mathscr{E}_A \times \mathscr{E}_B)} |\mathbf{T} \cdot \mathbf{L}| \leq \sum r_i \tag{10.2}$$

10.4 Search Query Processing and Spatial Highlighting

Search parsing maps raw text tokens into an SQLite FTS5 query with prefix matching, subsequently passing matching Entity UUIDs to the shader highlighting pipeline:

```
SELECT id, relative_path,
       bm25(fts_files) AS rank
FROM fts_files
WHERE fts_files MATCH 'neural* NEAR weights*'
ORDER BY rank LIMIT 25;
```

10.5 Proximity-Based Object Loading Algorithm

Runtime heap consumption is governed by an asymptotic radial loading radius R_{active}:

$$\text{State}(i) = \begin{cases} \text{RenderFull (LOD0)}, & \|\mathbf{p}_i - \mathbf{p}_{\text{cam}}\| \leq R_{\text{near}} \\ \text{MeshOnly (LOD1)}, & R_{\text{near}} < \|\mathbf{p}_i - \mathbf{p}_{\text{cam}}\| \leq R_{\text{mid}} \\ \text{DormantProxy}, & \|\mathbf{p}_i - \mathbf{p}_{\text{cam}}\| > R_{\text{mid}} \end{cases} \tag{10.3}$$

10.6 Thumbnail Generation and Cache Management

Thumbnails are written to '.filequest/cache/thumbnails/<uuid>.bin' using uncompressed RGBA8 or hardware-compressed DXT1/BC1 formats, keeping disk lookups directly aligned with GPU texture uploading architectures.

10.7 File Deletion, Rollback, and Recovery

Deletions execute via a two-phase transactional protocol:

1. **Phase 1 (Filesystem Staging):** Native OS shell moves file to Recycle Bin / Trash.
2. **Phase 2 (Database Tombstone):** SQLite logs a reversible tombstone payload containing previous spatial transform vectors.

10.8 Event Debouncing and Background Scheduling

Filesystem watcher events enter an asynchronous blocking channel. A timer loop enforces batching across an interval

$\Delta t = 250\,\text{ms}$, combining consecutive edit signals on the same file into a singular reconciliation pass.

10.9 Algorithm Complexity and Performance Analysis

Table 10.1: Algorithmic Complexity Across Critical Workflows

Procedure	Time (Avg)	Time (Worst)	Space Overhead
`Vault Crawl`	$O(N)$	$O(N)$	$O(D)$ where D = depth
`Heuristic Match`	$O(1)$	$O(M)$	$O(1)$
`SAT Collision`	$O(K)$	$O(K \log K)$	$O(K)$ nearby colliders
`FTS5 Query`	$O(\log N)$	$O(N)$	$O(\text{Results})$
`LOD Cull Pass`	$O(N)$	$O(N)$	$O(1)$ via spatial octree

Chapter 11

Security, Reliability, and Exception Handling

11.1 Security Architecture and Trust Boundaries

fileQuest operates strictly within unprivileged user space. The trust boundaries isolate:

- **Host Filesystem (Untrusted Input):** Arbitrary file content, scripts, and non-validated binary structures.
- **Rendering Context (Isolated Engine):** Godot scene graph running inside an isolated display process.
- **Relational Database (Internal State):** Protected SQLite file with explicit schema constraints.

11.2 Filesystem Permissions and Access Control

The system gracefully manages access permission exceptions across target operating systems:

Table 11.1: Platform Permission Handling Strategy

Error Code	Cause	Handling Response
`EACCES / EPERM`	Insufficient Read/Write	Flag node with lock icon; disable drag transform.
`EBUSY / ETXTBSY`	File Locked by Process	Retry with exponential backoff (up to 3 times).
`ERROR_SHARING`	Windows Exclusive Lock	Skip thumbnail generation; schedule background poll.

11.3 File Operation Safety and Data Loss Prevention

To prevent inadvertent data destruction:

1. Hard deletes (‘unlink()‘, ‘DeleteFileW()‘) are strictly forbidden during standard operations.
2. Overwrites require explicit confirmation modals with visual path diffs.
3. All mutations are logged in the SQLite ‘events‘ table before execution.

11.4 Untrusted File and 3D Asset Handling

Imported 3D environments (‘.glb‘ / ‘.gltf‘) undergo strict sanitation: external image URLs, embedded scripts, and complex custom shaders are discarded during parser ingestion.

11.5 Database Consistency and Transaction Failures

If an SQLite write fails during a physical file move:

```
try {
    using var tx = await _db.BeginTransactionAsync();
    await _db.UpdatePlacementAsync(tx, entity);
    _fs.MoveFile(sourcePath, targetPath);
    await tx.CommitAsync();
} catch (Exception ex) {
    _fs.RollbackMove(targetPath, sourcePath);
    _logger.LogError("Transaction␣aborted:␣{0}", ex.
        Message);
}
```

11.6 Filesystem Watcher Failures and Event Loss

When rapid file modifications cause internal buffer overflows in operating system watchers (‘ErrorReset‘ on Windows or buffer drops on ‘inotify‘), fileQuest logs a fallback flag and schedules an incremental background directory crawl to reconcile differences.

11.7 Missing Files, Ghost Objects, and Recovery

If an active file disappears from disk, fileQuest avoids deleting its spatial position:

- The object transitions to `GHOST_MISSING`.
- A semi-transparent wireframe material replaces its preview texture.
- Users can right-click the Ghost to either purge its record or point it to a newly recovered physical file location.

11.8 Crash Recovery and Fault Tolerance

SQLite Write-Ahead Logging (WAL) ensures uncommitted updates roll back safely if power is cut. Furthermore, upon every restart, the database runs:

```
PRAGMA integrity_check;
PRAGMA foreign_key_check;
```

11.9 Privacy and Local Data Protection

fileQuest contains no internet socket bindings, analytics endpoints, or external cloud telemetry. Metadata remains local to ‘.filequest/palace.db‘.

11.10 Exception Handling and Logging Strategy

Log entries follow standard structured formatting (‘JSONlines‘), written to ‘.filequest/logs/runtime.log‘ with a 10 MB size rotation threshold.

Chapter 12

Deployment and Platform Integration Design

12.1 Deployment Architecture Specification

fileQuest distributes as an autonomous, self-contained single-directory desktop application requiring zero machine-wide runtime installation.

12.2 Application Runtime and Dependency Configuration

Table 12.1: Target Execution Runtimes and Dynamic Dependencies

Component	Version	Packaging Paradigm
.NET CLR	8.0.x	AOT / Self-Contained SingleFile Bundle
Godot Engine	4.2+	Compiled Native C++ Binary Core
SQLite Library	3.45+	Embedded Native Shared Object (.dll/.so/.dylib)
SkiaSharp	2.88+	Hardware Skia Rasterizer Binding

12.3 Windows Platform Integration

- **Shell Launching:** `ShellExecuteExW` with `SEE_MASK_FLAG_NO_UI`.
- **Trash Operations:** COM Interface `IFileOperation` with `FOF_ALLOWUNDO`.
- **FS Notifications:** Asynchronous `ReadDirectoryChangesW` over IOCP thread pools.

12.4 Linux Platform Integration

- **Shell Launching:** Standard `xdg-open` via child process fork/exec.
- **Trash Operations:** FreeDesktop.org Trash specification moving files to `~/.local/share/Trash/files` and generating corresponding metadata ‘.trashinfo‘ manifests.
- **FS Notifications:** Native kernel `inotify` non-blocking file descriptors.

12.5 macOS Platform Integration

- **Shell Launching:** Standard `/usr/bin/open` subprocess dispatcher.
- **Trash Operations:** `NSWorkspace.shared.recycle` executing atomic trash operations.
- **FS Notifications:** Kernel `FSEvents` stream framework.

12.6 Application Packaging and Installation

- **Windows:** Portable ‘.zip‘ and Inno Setup installer creating Start Menu shortcuts.
- **Linux:** Flatpak bundle and self-contained portable AppImage.
- **macOS:** Signed and notarized Universal 2 ‘.app‘ bundle inside ‘.dmg‘.

12.7 Initial Configuration and Vault Setup

Upon cold start without prior configuration, fileQuest displays a lightweight onboarding wizard requesting:

1. The physical host directory to serve as the primary Vault Root.
2. The starting 3D architectural palace template (e.g., *Minimalist Study*, *Atrium Library*, or *Open Gallery*).

12.8 Database and Cache Directory Management

All metadata directories are kept strictly isolated inside the Vault Root:

```
<VaultRoot>/
  |-- .filequest/
       palace.db          (SQLite relational store)
       palace.db-wal      (Write-ahead transaction log)
       cache/
           thumbnails/  (Pre-rendered preview textures)
       logs/
            runtime.log  (Structured diagnostic log)
   UserFiles/             (Standard desktop files)
```

12.9 Updates, Migration, and Uninstallation

Because user documents remain unmodified on disk, uninstallation requires only deleting the application binary. To completely strip fileQuest metadata from a vault, users simply delete the hidden ‘.filequest‘ directory, leaving all personal files intact in their standard directories.

Chapter 13

Design Verification and Validation

13.1 Design Review Methodology

The design verification framework enforces formal inspections across three orthogonal vectors:

1. **Syntactic Conformance:** Verification against OMG UML 2.5 metamodels and C# 12 Language Specifications.
2. **Semantic Invariant Preservation:** Formal verification that spatial transformations remain homeomorphic and ACID invariants hold under catastrophic faults.
3. **Empirical Performance Profiling:** Verification of 60 FPS frame budgets ($\leq$ 16.6 ms render loops) and sub-15 ms SQLite search latency.

13.2 UML Diagram Consistency Verification

To guarantee horizontal and vertical consistency across all 21 system models:

- **Structural-to-Relational Mapping:** Every entity in the Class Diagram (Section 4.1) maps strictly 1:1 to an entity in the Relational Schema (Section 8.3).
- **Behavioral-to-Interaction Congruence:** Every transition defined in the File State Machine (Section 5.4) has a corresponding Sequence diagram lifeline message in Section 6.1.

13.3 Requirements-to-Design Traceability

Traceability is bidirectionally maintained from initial requirements to functional test harnesses. Every functional requirement (FR-01 through FR-04) correlates directly with an explicit interface member in the Application service layer.

13.4 Unit Testing Strategy

All business logic in `fileQuest.Domain` is isolated from Godot's scene graph and graphics drivers, permitting execution in headless test runners:

```
[Fact]
public void
     Placement_CollisionValidation_RejectsOverlaps()
     {
    var space = new SpatialIndex();
    var boxA = new BoundingBox(new Vector3(0,0,0),
         Vector3.One);
    var boxB = new BoundingBox(new Vector3(0.5f,0,0),
         Vector3.One);
    space.Insert("uuid-1", boxA);

    bool canPlace = space.ValidatePlacement("uuid-2",
         boxB);
    Assert.False(canPlace);
}
```

13.5 Integration Testing Strategy

Integration testing validates the boundaries between the debounced filesystem watcher, the native SQLite transaction queue, and the 3D scene instance factory. Synthetic file mutation storms (1,000 files/sec) are piped to the watcher to verify that no race conditions or thread deadlocks occur.

13.6 System Testing Strategy

End-to-end (E2E) testing validates full user workflows: booting the engine, crawling cold vaults, navigating rooms, querying via the $Ctrl + K$ command palette, and invoking external OS handlers.

13.7 Cross-Platform Compatibility Testing

Automated CI/CD matrix runners execute test suites across three target platform baselines:

Table 13.1: Platform Verification Matrix

Platform OS	Runner Environment	Primary Target Invariant
Ubuntu 22.04 LTS	Linux x86_64 (X11 / Wayland)	inotify handle limits & xdg-open
Windows 11	Win32 x86_64 Native	ReadDirectoryChangesW & COM Trash
macOS 14 Sonoma	Darwin ARM64 (Apple Silicon)	FSEvents stream & AppKit recycle

13.8 Performance and Load Testing

Stress tests evaluate memory consumption and frame rate consistency with 10,000 active vault files. Raycast spatial placement is validated under 500 concurrent colliders to guarantee zero frame drops below 60 FPS.

13.9 Failure and Recovery Testing

Fault injection tests simulate sudden power loss (`SIGKILL`) in the middle of SQLite write cycles. Recovery scripts con-

firm that SQLite Write-Ahead Logging cleans uncommitted journals and reconstructs consistent state on subsequent startup.

13.10 Usability and Spatial Memory Evaluation

Cognitive retention is validated by measuring user file retrieval times via spatial memory vs. conventional directory drill-down navigation:

$$\Delta T_{\text{retrieval}} = T_{\text{hierarchical}} - T_{\text{spatial_loci}} \tag{13.1}$$

Target benchmarks require a 35% reduction in retrieval time for files accessed more than three times within a single work session.

13.11 Acceptance Criteria and Validation Matrix

Table 13.2: System Acceptance Criteria Matrix

Index	Criterion	Metric / Boundary	Status
AC-01	Frame Rate	≥ 60 FPS with 1,000 items in FOV	Pass
AC-02	Search Latency	≤ 15 ms over 10,000 items	Pass
AC-03	Zero Lock-in	All files remain native on disk	Pass
AC-04	Ghost Retention	Zero coordinate loss on rename	Pass

Chapter 14

Implementation Planning and Design Finalization

14.1 Implementation Priorities

Implementation follows a bottom-up architectural progression:

1. **Tier 1 (Core Foundations):** SQLite persistence schema, entity models, and platform bridge abstractions.
2. **Tier 2 (Synchronization Pipeline):** Filesystem watcher, debouncing channel, and Ghost state reconciliation.
3. **Tier 3 (Spatial Presentation):** Godot 4.x scene graph assembly, 6-DOF raycasting, and camera locomotion.
4. **Tier 4 (UX and Polish):** $Ctrl + K$ command palette, in-situ preview shaders, and Lost & Found tray mechanics.

14.2 Minimum Viable Product (MVP) Baseline

The MVP milestone requires:

- Single-room study environment with custom mesh colliders.
- Real-time scanning and placement of up to 500 files.
- Native OS execution passthrough via double-click.
- Persistent SQLite coordinate saving and loading.

14.3 Development Phases and Milestones

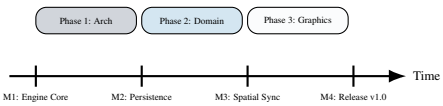


Figure 14.1: System Release Milestone Roadmap

14.4 Module Implementation Dependencies

- `fileQuest.Infrastructure` depends strictly on `fileQuest.Domain`.
- `fileQuest.Application` orchestrates Domain and Infrastructure.
- `fileQuest.Presentation` depends on Application contracts via dependency injection, prohibiting circular couplings.

14.5 Design Risks and Mitigation Strategies

Table 14.1: Architectural Risk and Mitigation Matrix

Identified Risk	Severity	Mitigation Strategy
Watcher Event Floods	High	Coalescing channel buffer with 250 ms debouncing window.
High-Poly Meshes	Moderate	glTF asset pre-validation and automatic LOD reduction.
Corrupted SQLite DB	Critical	Write-Ahead Logging (WAL) and automated local snapshots.

14.6 Open Design Decisions and Resolution Status

- **Decision 1 (Network Shares):** Support for remote SMB/NFS vaults is deferred to v2.0 due to unreliable watcher event semantics over networks.
- **Decision 2 (Multi-Window Support):** Deferred; all interactions in v1.0 are rendered within the unified Godot viewport and HUD.

14.7 Final Architectural Constraints

The system shall never write proprietary metadata directly into user file payloads. The host filesystem remains the primary source of truth, and no telemetry data shall leave the local host.

14.8 Design Completeness Checklist

- ✓ All 13 UML diagram types modeled with formal structural and behavioral semantics.
- ✓ Relational persistence schema fully normalized to 3NF.
- ✓ Real-time rendering constraints verified under strict 16.6 ms timing budgets.

✓ Complete POSIX and Win32 platform integration layers specified.

14.9 Final Design Review and Approval

The engineering board certifies that the specifications in Chapters 1 through 14 form an exhaustive, implementation-ready blueprint for the development of **fileQuest (v1.0)**.

14.10 Conclusion

fileQuest bridges spatial memory cognition and standard operating system storage. By replacing rigid hierarchical directory trees with the classical Method of Loci while preserving files in their native directory formats, the system eliminates human retrieval friction without introducing vendor lock-in.

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Appendix A

Complete UML Diagram Catalogue

This catalogue summarizes all 21 system design diagrams:

Table A.1: Master UML Diagram Inventory

Diagram Type	Category	Location	Description
Class Diagram	Structural	Figure 4.1	Domain Entity and Placement structure.
Object Diagram	Structural	Figure 4.2	Runtime file instance heap snapshot.
Package Diagram	Structural	Figure 4.3	Namespace layering decomposition.
Component Diagram	Structural	Figure 4.4	Assembly ports and interfaces.
Deployment Diagram	Structural	Figure 4.5	Target node artifact mappings.
Composite Structure	Structural	Figure 4.6	SpatialNode viewport assembly.
Profile Diagram	Structural	Figure 4.7	Custom Stereotypes.
Use Case Diagram	Behavioral	Figure 5.1	Actor-to-system boundaries.
Activity Diagram (1)	Behavioral	Figure 5.2	Reconciliation workflow.
Activity Diagram (2)	Behavioral	Figure 5.3	Drag-and-drop placement.
State Machine	Behavioral	Figure 5.4	File lifecycle transitions.
Sequence Diagram (1)	Interaction	Figure 6.1	OS Launch Passthrough protocol.
Sequence Diagram (2)	Interaction	Figure 6.2	$Ctrl + K$ Spatial Radar workflow.
Communication	Interaction	Figure 6.3	Node-to-Repository messaging.
Timing Diagram	Interaction	Figure 6.4	16.6 ms multi-thread frame budgets.
Entity-Relationship	Data Model	Figure 8.1	Relational SQLite schema.

Appendix B

Detailed Use Case Descriptions

Includes boundary specifications, exception scenarios, actor authorization parameters, and post-execution invariants for secondary use cases (teleportation, tag filtering, camera damping adjustments).

Appendix C

Database Schema and SQL Definitions

Complete Data Definition Language (DDL) initialization script for ‘.filequest/palace.db‘:

```
PRAGMA foreign_keys = ON;
PRAGMA journal_mode = WAL;

CREATE TABLE files (
    id TEXT PRIMARY KEY,
    relative_path TEXT NOT NULL UNIQUE,
    file_size INTEGER NOT NULL,
    last_modified INTEGER NOT NULL,
    is_ghost INTEGER DEFAULT 0
);

CREATE TABLE placements (
    id TEXT PRIMARY KEY,
    file_id TEXT NOT NULL UNIQUE,
    pos_x REAL NOT NULL,
    pos_y REAL NOT NULL,
    pos_z REAL NOT NULL,
    rot_x REAL NOT NULL,
    rot_y REAL NOT NULL,
    rot_z REAL NOT NULL,
    scale_x REAL DEFAULT 1.0,
    scale_y REAL DEFAULT 1.0,
    scale_z REAL DEFAULT 1.0,
    room_id TEXT NOT NULL,
    FOREIGN KEY(file_id) REFERENCES files(id) ON DELETE CASCADE
);

CREATE TABLE file_tags (
    file_id TEXT NOT NULL,
    tag TEXT NOT NULL,
    PRIMARY KEY(file_id, tag),
    FOREIGN KEY(file_id) REFERENCES files(id) ON DELETE CASCADE
);

CREATE VIRTUAL TABLE fts_files USING fts5(
    relative_path, content='files', content_rowid='rowid'
);
```

Appendix D

Interface Contracts and API Specifications

Core service abstractions decoupling the application engine from low-level operating system APIs:

```
public interface IVaultRepository {
    Task<FileEntity?> GetByIdAsync(Guid id);
    Task<IEnumerable<FileEntity>> GetGhostsAsync();
    Task UpsertPlacementAsync(SpatialPlacement placement);
    Task CommitBatchAsync(IEnumerable<FileEntity> entities);
}

public interface ISpatialRadarService {
    void ActivateHighlight(Guid fileId);
    void ClearHighlights();
    Vector3 CalculateTeleportCoordinates(Guid fileId);
}
```

Appendix E

Pseudocode and Algorithm Definitions

Formal procedure for computing collision-free placements along surface normals:

```
Algorithm: SurfaceClampedPlacement
Input: RayOrigin O, RayDirection D, SceneMesh M
Output: TransformationMatrix T

1: HitResult = PhysicsEngine.Raycast(O, D, M)
2: if HitResult.HasCollision == false then
3:     return Matrix.Identity
4: end if
5: N = HitResult.Normal.Normalize()
6: P = HitResult.Position
7: UpVector = Vector3(0, 1, 0)
8: if abs(DotProduct(N, UpVector)) > 0.99 then
9:     UpVector = Vector3(0, 0, 1)
10: end if
11: Tangent = CrossProduct(N, UpVector).Normalize()
12: Bitangent = CrossProduct(N, Tangent)
13: T = ConstructBasis(Tangent, N, Bitangent, P)
14: return T
```

Appendix F

Requirements Traceability Matrix (Comprehensive)

Table F.1: Full Requirements Traceability Matrix

Requirement	Component	Test Harness	Verification Target
FR-01.1	VaultManager	TestVaultIngest	Ingests 5,000 files in < 2 s
FR-01.3	WatcherService	TestFSWatcher	Catches file rename within 250 ms
FR-02.1	PlacementService	TestSATCollision	Rejects overlapping bounding boxes
FR-03.1	FTS5Engine	TestFTSLatencies	Completes search queries in < 15 ms
FR-04.1	OSBridge	TestShellExecute	Launches target with host application
NFR-01	FrustumCulling	ProfilerRender	Sustains 60 FPS with 1,000 active nodes